

1 Which of the following gives the correct base units for the permittivity of free space, ϵ_0 ?

A $\text{C}^2 \text{N}^{-1} \text{m}^{-2}$

B $\text{kg}^{-1} \text{m}^{-3} \text{s}^4 \text{A}^2$

C $\text{kg} \text{m}^{-3} \text{A}^2$

D $\text{kg}^{-1} \text{m}^3 \text{s}^2 \text{A}^{-2}$

1 **B** $\epsilon_0 = \frac{Qq}{4\pi Fr^2}$ from Electric Force Formula.

Q, q have units A. s

F has units kgms^{-2}

r has unit as m

Therefore, ϵ_0 has units $\text{kg}^{-1} \text{m}^{-3} \text{s}^4 \text{A}^2$.

2 A photographer wishes to check the time for which the shutter on a camera stays